PASSBOOK · CA INTERMEDIATE · GROUP II

Paper 6A

Financial Management

Revision Cards

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CHAPTER 1

Scope and Objectives of Financial Management

1.1 Objectives: profit maximisation, wealth maximisation and value creation

1 **What are the four limitations of profit maximisation as an objective, in ICAI's order?**

A (i) The term profit is vague; (ii) it ignores risk; (iii) it does not take into account the time pattern of returns; (iv) it is too narrow (ignores social considerations and obligations).

2 **How is shareholders' wealth defined in the wealth maximisation model?**

A Wealth = present value of benefits − present value of costs: cost-benefit analysis adjusted for timing and risk (time value of money).

3 **Which three things does the finance manager follow to measure and maximise shareholders' wealth?**

A Cash flow approach (not accounting profit); cost-benefit analysis; application of the time value of money.

4 **According to Van Horne, what represents the value of a firm?**

A The market price of the company's common stock: the focal judgment of all market participants, a performance index or report card of the firm's progress.

5 **Write the two formulas for the value of a firm given in the Study Material.**

A V = N × MP (number of shares × market price of shares), or V = Ve + Vd (value of equity + value of debt).

6 **What does the shareholder value maximisation model hold?**

A That the primary goal of the firm is to maximise its market value; business decisions should seek to increase the net present value of the economic profits of the firm.

7 **Profit maximisation: the objective, the two advantages and the four disadvantages in ICAI's table?**

A Objective: large amount of profits. Advantages: easy to calculate profits; easy to determine the link between financial decisions and profits. Disadvantages: emphasises short-term gains; ignores risk or uncertainty; ignores timing of returns; requires immediate resources.

8 **Wealth maximisation: the objective, the four advantages and the two disadvantages in ICAI's table?**

A Objective: highest market value of shares. Advantages: long-term gains; recognises risk or uncertainty; recognises timing of returns; considers shareholders' return. Disadvantages: no clear relationship between financial decisions and share price; can lead to management anxiety and frustration.

9 **When do profit maximisation and wealth maximisation amount to essentially the same?**

A Where the time period is short and the degree of uncertainty is not great.

10 **Why does wealth maximisation remain the primary goal over other goals such as growth or community welfare?**

A It is critical for the very existence of the business: if it is not met, the public and institutions lose confidence and will not invest further in its growth, and restricted growth leaves the other goals unfulfilled.

11 **Value creation: ICAI's one-line meaning in the MTP Suggested Answer?**

A Making strategic decisions that generate sustainable returns exceeding the cost of capital.

1.2 The finance function, the financial manager and the controller

12 **What are the four stages of decision making for a new venture in the Study Material?**

A (1) Decide which assets to buy; (2) determine the total investment needed for them; (3) determine the cash needed to run daily operations (working capital requirement); (4) decide the sources to finance the total investment.

13 **Define financial management (ICAI's first meaning).**

A That managerial activity which is concerned with planning and controlling of the firm's financial resources: acquiring, financing and managing assets to accomplish the overall goal of a business enterprise (mainly to maximise shareholders' wealth).

14 **What are the two basic aspects of financial management?**

A Procurement of funds, and effective utilisation of those funds to achieve business objectives.

15 **In which three characteristics do funds from different sources differ?**

A Risk, cost and control.

16 **Why is equity the best source from the risk point of view but usually the most expensive?**

A No repayment except in liquidation; but shareholders expect more than the prevalent interest rate, dividends are an appropriation of profit not allowed as an expense under the Income Tax Act, and a new issue may dilute control.

17 **State the expression for the value of a firm in terms of finance decisions, and name the decisions.**

A V = f(I, F, D): investment, financing and dividend decisions (long term); short term, working capital management.

18 **Which part of the dividend decision is also a financing decision?**

A The amount retained to support the growth of the organisation.

19 **Ezra Solomon's four aspects of the scope of financial management?**

A Size and rate of growth of the enterprise; composition of its assets; mix of its financing (debt to equity); analysis, planning and control of its financial affairs.

20 **What are the three phases in the evolution of financial management?**

A Traditional (needed only for occasional events such as mergers and liquidation), transitional (day-to-day problems of funds analysis, planning and control), modern (still going on; financial analysis and theories).

21 **The five responsibilities of the finance executive?**

A Financial analysis and planning; investment decisions; financing and capital structure decisions; management of financial resources (such as working capital); risk management.

22 **Which managers report to the treasurer, and which to the controller, in ICAI's organisation chart?**

A Treasurer: cash, credit, capital budgeting, fund raising and portfolio managers. Controller: financial accounting, cost accounting, tax and data processing managers, and the internal auditor.

23 **How has the role of the financial controller changed?**

A Once limited to procurement of funds for major events (promotion, expansion, merger); now also the investment, financing and dividend decisions, maximising returns in relation to the given risk.

1.3 Agency problem, financial

environment, distress and insolvency

24 **Define the agency problem (ICAI's "in a nutshell" line).**

A The chances that managers may place personal goals ahead of the goal of owners.

25 **Why does the agency problem arise in companies but not in a sole proprietorship?**

A In a proprietorship or partnership the owners participate in management; in a company the owners are not active in management, so owners and managers are separate (a principal–agent relationship).

26 **Define agency cost.**

A The additional cost borne by the shareholders to monitor the manager and control their behaviour so as to maximise shareholders' wealth.

27 **What are the four types of agency cost, in ICAI's order?**

A Monitoring, bonding, opportunity, structuring.

28 **How is the agency problem of debt lenders addressed?**

A By imposing negative covenants, i.e. the managers cannot borrow beyond a point.

29 **What three efforts does the Study Material list for addressing the agency problem between managers and shareholders?**

A Managerial compensation linked to profit and to long-term objectives; employee schemes based on the assumption that maximising the stock price is the investors' objective; effective monitoring.

30 **Define financial distress.**

A A position where cash inflows of a firm are inadequate to meet all its current obligations.

31 **Define insolvency, and state how it relates to distress.**

A The inability of a firm to repay various debts; it is a result of continuous financial distress.

32 **What happens if financial distress continues for a long period?**

A The firm may have to sell its assets, many times at a lower price; when revenue is inadequate to revive the situation, it cannot meet its obligations and becomes insolvent.

33 **Financial management versus accounting: how do they differ in the treatment of funds?**

A Accounting measures funds on the accrual principle (revenue at sale, expenses when incurred); financial management on cash flows (revenue when received, expenses when paid), to maintain solvency.

34 **Which discipline is primary for the financial manager, and which are supportive?**

A Accounting is primary; marketing, production and quantitative methods are supportive; economics helps the manager know the world outside the business.

CHAPTER 2

Financial Analysis and Planning: Ratio Analysis

2.1 Liquidity and capital structure ratios

1 **ICAI's definition of a ratio?**

A "The indicated quotient of two mathematical expressions and as the relationship between two or more things."

2 **Formula: quick (acid-test) ratio, and what is excluded from quick assets?**

A Quick assets ÷ current liabilities; quick assets = current assets − inventories − prepaid expenses. Norm 1:1.

3 **What is the generally acceptable current ratio, and what does ICAI say about it?**

A 2:1; whether a specific ratio is satisfactory depends on the nature of the business and the characteristics of its current assets and liabilities.

4 **Formula: basic defense interval?**

A (Cash and bank + net receivables + marketable securities) ÷ daily operating expenses, where daily operating expenses = (COGS + selling, administration and general expenses − depreciation and other non-cash expenditure) ÷ days in the year.

5 **Formula: cash (absolute liquidity) ratio?**

A (Cash and bank balances + marketable securities, or current investments) ÷ current liabilities.

6 **Formula: capital gearing ratio?**

A (Preference share capital + debentures + other borrowed funds) ÷ (equity share capital + reserves and surplus − losses).

7 **Formula: proprietary ratio, and what is in proprietary fund?**

A Proprietary fund ÷ total assets; proprietary fund = equity share capital + preference share capital + reserves and surplus (total assets exclude fictitious assets and losses).

8 **Formula: equity ratio and debt ratio?**

A Equity ratio = shareholders' equity ÷ net assets; debt ratio = total debt ÷ net assets (net assets = net fixed assets + net current assets).

9 **Formula: debt service coverage ratio, and ICAI's satisfactory range?**

A (EAT + depreciation and other non-cash expenses + interest + adjustments such as loss on sale of fixed assets) ÷ (interest + instalments). Normally 1.5 to 2 is satisfactory; the summary table gives 2 as ideal.

10 **Why is EBIT, not PAT, used in the interest coverage ratio?**

A Interest on debt is a deductible expense, so the ability to pay interest is not affected by the tax burden.

11 **Formula: fixed charges coverage ratio?**

A (EBIT + depreciation) ÷ (interest + repayment of loan); more than 1 is considered safe.

12 **Which ratio does ICAI call the indicator of a firm's financial leverage?**

A The debt to equity ratio.

2.2 Activity (turnover) ratios

13 **Why does ICAI call activity ratios "asset management ratios"?**

A Because they evaluate the efficiency with which the firm manages and utilises its assets.

14 **List ICAI's activity ratios in order.**

A Total assets turnover; fixed assets turnover; capital (net assets) turnover; current assets turnover; working capital turnover, segregated into inventory, receivables and payables turnover.

15 **Formula: inventory turnover ratio and average inventory?**

A Cost of goods sold (or sales) ÷ average inventory; average inventory = (opening stock + closing stock) ÷ 2.

16 **Formula: raw material inventory turnover ratio?**

A Raw material consumed ÷ average raw material stock.

17 **Formula: average collection period, two ways?**

A Average accounts receivable ÷ average daily credit sales; or 360 days (12 months, 52 weeks) ÷ receivables turnover ratio.

18 **Formula: payables turnover ratio?**

A Annual net credit purchases ÷ average accounts payable.

19 **What does a very high working capital turnover ratio indicate, per ICAI?**

A That the company needs to raise additional working capital for future needs.

20 **What differentiates cost of goods sold from cost of sales?**

A Only selling and distribution expenses; in their absence COGS equals cost of sales.

21 **Why may a firm with old plant show a higher fixed assets turnover?**

A Its plant stands at a lower written-down value, so the same sales are divided by a smaller fixed asset figure.

22 **How are purchases found from a trading account?**

A Purchases = cost of goods sold + closing stock − opening stock.

23 **What does a low debtors' turnover ratio reflect?**

A Liberal credit terms granted to customers (a high ratio shows rapid collection).

2.3 Profitability and market ratios, and the DuPont analysis

24 **ICAI's four groups of profitability ratios?**

A (i) Related to sales; (ii) related to overall return on assets / investments; (iii) required for analysis from the owner's point of view; (iv) related to market / valuation / investors.

25 **Formula: operating ratio, and how it relates to the operating profit ratio?**

A (COGS + operating expenses) ÷ sales × 100; operating ratio + operating profit ratio = 100% when operating profit = sales − COGS − operating expenses.

26 **ROI split into two ratios?**

A ROI = (return ÷ sales) × (sales ÷ investment) = profitability ratio × investment turnover ratio.

27 **Formula: ROCE pre-tax and post-tax, and three ways to find capital employed?**

A EBIT ÷ capital employed × 100; EBIT(1 – t) ÷ capital employed × 100. Capital employed = total assets – current liabilities = fixed assets + working capital = equity + long-term debt.

28 **ICAI's rule for ROCE against the borrowing rate?**

A ROCE should always be higher than the rate at which the company borrows.

29 **Formula: return on equity?**

A (Net profit after taxes – preference dividend, if any) ÷ net worth (equity shareholders' funds) × 100.

30 **DuPont model: the three components of ROE?**

A Net profit margin (net income ÷ sales) × asset turnover (sales ÷ assets) × equity multiplier (assets ÷ shareholders' equity).

31 **What does the equity multiplier show in the DuPont model?**

A Financial leverage: what portion of the return on equity is the result of debt.

32 **Formula: dividend pay-out ratio?**

A Dividend per equity share (DPS) ÷ earnings per share (EPS).

33 **What does a high P/E ratio indicate, per ICAI?**

A Either the share is over-valued or investors expect high growth rates in future; the P/E also indicates the payback period to investors.

34 **Two formulas for dividend yield?**

A DPS ÷ MPS × 100; or (dividend ± change in share price) ÷ initial share price × 100.

35 **Q ratio: who proposed it, the formula, and the equilibrium value?**

A James Tobin; market value of equity and liabilities ÷ estimated replacement cost of assets; equilibrium at 1 (below 1 possibly undervalued, above 1 possibly overvalued).

2.4 Building financial statements

from ratios; limitations of ratios

36 **ICAI's eight limitations of financial ratios, in order?**

A Diversified product lines; distortion by inflation; seasonal factors; window dressing of popular ratios; differences in accounting policies and accounting period; no standard set of ratios to compare with; difficulty to generalise whether a ratio is good or bad; ratios are inter-related, not independent.

37 **What closing line does ICAI give after the limitations of ratios?**

A Financial ratios provide clues but not conclusions; they are tools only in the hands of experts, as there is no standard ready-made interpretation of financial ratios.

38 **What is window dressing, in ICAI's limitation?**

A Year-end adjustments made to give a good shape to popular ratios such as the current ratio and debt-equity ratio, changing the character of the ratios.

39 **How does ICAI suggest dealing with the interdependence of ratios?**

A Through multivariate analysis: analysing the relationship between several variables simultaneously.

40 **Horizontal vs vertical analysis?**

A Horizontal: one year's statements compared with another year or years. Vertical: a single year analysed, P&L items as % of gross sales and balance sheet items as % of total assets; useful for inter-firm comparison.

41 **The six aspects of a firm's performance for which ratio analysis is relevant?**

A Liquidity position; long-term solvency; operating efficiency; overall profitability; inter-firm comparison; financial ratios for budgeting.

42 **Current liabilities from working capital and current ratio?**

A CL = working capital ÷ (current ratio − 1); CA = CL × current ratio.

43 **Closing stock from average stock and opening stock?**

A Closing stock = 2 × average stock − opening stock.

44 **Gross profit of g% on cost as a percentage of sales?**

A g ÷ (100 + g) × 100; e.g. 25% on cost = 20% of sales.

45 **Which ratios does the Study Material say a company must now disclose in the notes to accounts?**

A Current ratio, debt-equity, DSCR, ROE, inventory turnover, trade receivables turnover, trade payables turnover, net capital turnover, net profit ratio, ROCE and return on investment.

46 **When must you let sales be x in a ratio question?**

A When the only rupee figure (for example cash) enters the balance sheet only through the totals: write every item in terms of x and solve total assets = total liabilities.

CHAPTER 3

Types of Financing

3.1 Long-term sources: equity, preference, debt and retained earnings

1 **What is the basic principle for funding various financial needs?**

A Short-term needs from short-term sources, medium-term needs from medium-term sources and long-term needs from long-term sources; the method of raising funds is decided by the period for which the funds are required.

2 **State four advantages of raising funds by issue of equity shares.**

A (i) A permanent source: no liability for redemption; (ii) increases the financial base and borrowing powers; (iii) no legal obligation to pay dividends; (iv) share capital can be further increased by a rights issue.

3 **State three disadvantages of raising funds by equity shares.**

A (i) Investors find them riskier (uncertain dividends and capital gains); (ii) a new issue reduces the EPS of existing shareholders until profits rise proportionately; (iii) a new issue can reduce the ownership and control of existing shareholders.

4 **Why is preference share capital called a hybrid form of financing?**

A Like equity, its dividend is not a tax-deductible payment; like debt, its rate of dividend is fixed.

5 **Name the six types of preference shares in the Study Material.**

A Cumulative, non-cumulative, redeemable, participating, non-participating and convertible.

6 **State the two disadvantages of preference shares.**

A (i) The dividend is not tax-deductible, so no tax shield: costlier than debt; (ii) dividends are cumulative, and until arrears are paid no dividend can be paid to equity shareholders, which could impair the company's reputation.

7 **State three advantages of raising finance by debentures.**

A (i) Cost much lower than preference or equity, as interest is tax-deductible and investors accept a lower return; (ii) no dilution of control; (iii) advantageous in a period of rising prices, as the fixed monetary outgo falls in real terms.

8 **State four disadvantages of debenture financing.**

A (i) Interest and principal are obligatory payments; (ii) protective covenants may be restrictive; (iii) it enhances the financial risk of the firm; (iv) a large cash outflow is needed at maturity.

9 **Callable bond versus puttable bond?**

A Callable: the issuer has the right to redeem before maturity at a predetermined call price (generally at a premium). Puttable: the investor has the right to sell the bond back to the company before maturity.

10 **What is a masala bond?**

A A rupee-denominated bond that Indian corporate borrowers sell to investors in overseas markets: issued outside India but denominated in Indian rupees.

11 **State the key features of bridge finance.**

A (i) A short-period loan, normally from commercial banks, pending disbursement of loans sanctioned by financial institutions; (ii) repaid or adjusted out of the term loans when disbursed; (iii) normally secured by hypothecating movable assets, personal guarantees and demand promissory notes; (iv) interest generally higher than on term loans.

12 **What is a working capital term loan (WCTL)?**

A Long-term bank funding of the portion of working capital that is always required (the minimum level) and is not affected by seasonal requirements.

3.2 Venture capital, lease financing, sale and lease back and other sources

13 **What is venture capital financing?**

A Financing of new high risky ventures promoted by qualified entrepreneurs who lack experience and funds to give shape to their ideas; the venture capitalist buys equity or debt securities of such ventures with potential to succeed in future.

14 **State three characteristics of venture capital financing.**

A (i) Basically equity finance in new companies; (ii) a long-term investment in growth-oriented small/medium firms; (iii) besides funds, support in sales strategy, business networking and management expertise.

15 **Name the four methods of venture capital financing.**

A Equity financing (contribution not exceeding 49% of total equity), conditional loan (royalty, no interest), income note (interest and royalty at low rates) and participating debenture (no, low, then high interest in three phases).

16 **What is a conditional loan in venture capital financing?**

A A loan repayable as a royalty (2–15% in India) after the venture generates sales, with no interest; some financiers allow a high rate of interest instead of royalty once the venture is commercially sound.

17 **What is debt securitisation?**

A A process in which illiquid assets are pooled into marketable securities that can be sold to investors, creating instruments that represent ownership interest in, or are secured by, a segregated income-producing pool of assets.

18 **Name the three functions in the debt securitisation process.**

A The origination function, the pooling function (transfer of the pool to an SPV acting as trustee for investors) and the securitisation function (the SPV issues securities against the pool).

19 **Financial lease versus operating lease: who bears the risk of obsolescence?**

A Financial lease: the lessee. Operating lease: the lessor.

20 **What is a leveraged lease?**

A A lease with a third party besides lessor and lessee: the lessor borrows part of the cost (say 80%) from a lender against the asset; the lessee pays the lender directly from the rentals and the surplus goes to the lessor, who claims depreciation.

21 **What is a sale and lease back?**

A The owner sells an asset to a buyer who leases the same asset back to the owner for lease rentals; the asset is not physically exchanged; the seller becomes the lessee and gets the price, the buyer the lessor and gets the rentals.

22 **What is a sales-aid lease?**

A A lease where the lessor ties up with a manufacturer to market its product through its leasing operations; the manufacturer grants the lessor credit or a commission, so the lessor earns from both lessee and manufacturer.

23 **State the key features of seed capital assistance.**

A An IDBI scheme for qualified or experienced entrepreneurs lacking funds; interest free with a service charge of 1% p.a. for the first five years and increasing thereafter (IDBI may charge interest if the company can pay); repayment fixed by repaying capacity, with an initial moratorium up to five years.

24 **What are junk bonds, and why are they called high-yield bonds?**

A Bonds rated below investment grade by credit rating agencies, with a low credit rating and high risk of default; investors are compensated with higher interest rates, hence high-yield bonds.

3.3 Short-term sources of finance

25 **State the features of trade credit.**

A Credit granted by suppliers as an incident of sale; usually 15 to 90 days; generates automatically and is common to almost all businesses; open account or bills payable; without any explicit cost and keeps rotating while the business is a going concern; enhances automatically with the volume of business.

26 **Why are accrued expenses and deferred income called spontaneous sources of finance?**

A Accrued expenses (wages, taxes, interest, dividends for services already received) arise out of day-to-day activities; deferred income (money received for goods and services to be provided in future) increases liquidity. Both arise without negotiation in the course of business.

27 **Explain in brief the features of commercial paper.**

A An unsecured money market instrument issued as a promissory note; RBI scheme of 1989 for highly rated corporate borrowers (also primary dealers and all-India financial institutions); denominations of ₹5 lakh or multiples; interest generally linked to the one-year government bond yield; credit rating from CRISIL, ICRA, CARE, FITCH or another RBI-specified agency.

28 **What is a certificate of deposit, and its main advantage?**

A A document of title like a time deposit receipt issued by a bank, maturity not less than 7 days up to one year, with no prescribed interest rate. The banker need not encash it before maturity, and the investor has liquidity by selling it in the secondary market.

29 **Treasury bills: issuer and maturity?**

A A class of Central Government securities issued by the Government of India for short-term borrowing, with maturities of 14 to 364 days.

30 **Cash credit versus overdraft?**

A Overdraft: withdrawal beyond the current account balance up to a limit, interest on daily balances, repayable on demand but renewed annually. Cash credit: a limit drawn as required, interest only on the amount availed, generally against tradable goods by pledge or hypothecation.

31 **What is a clean overdraft?**

A An advance relying on the personal security of a financially sound borrower of known integrity, with no tangible security; banks look at turnover and dealings, take guarantees of creditworthy persons, and grant it for a short period only.

32 **What is packing credit, and within what period must it be liquidated?**

A A pre-shipment advance by a bank to an exporter with a firm export order or irrevocable letter of credit, for buying, manufacturing, processing, packing and shipping goods; liquidated within 180 days from its commencement by negotiation of export bills or receipt of export proceeds.

33 **Name the five types of packing credit.**

A Clean packing credit; packing credit against hypothecation of goods; packing credit against pledge of goods; ECGC guarantee; forward exchange contract.

34 **Name the four forms of post-shipment finance.**

A Purchase/discounting of documentary export bills; ECGC guarantee; advance against export bills sent for collection; advance against duty drawbacks, cash subsidy, etc.

35 **What are inter-corporate deposits?**

A Short-period borrowings, say 6 months, from other companies with surplus liquidity, at a rate that varies with the amount and the period.

36 **Public deposits: limit, period and use, as the Study Material states them?**

A Up to 35% of paid-up capital and reserves, subject to RBI stipulations; for six months to three years; unsecured; not to be used for fixed assets, mainly for working capital.

3.4 Contemporary and

international sources of finance

37 **What are external commercial borrowings (ECBs)?**

A Commercial loans (bank loans, buyers' credit, suppliers' credit, securitised instruments such as FRNs and fixed rate bonds) availed from non-resident lenders with a minimum average maturity of 3 years.

38 **Through which routes can ECBs be accessed?**

A The automatic route (no RBI or Government approval; companies registered under the Companies Act, NGOs in micro finance) and the approval route (approval needed; e.g. FIs and banks dealing exclusively in infrastructure or export finance).

39 **Euro bond versus foreign bond?**

A Euro bond: not denominated in the currency of the country where it is issued (a yen note floated in Germany), often bearer. Foreign bond: issued by a foreign corporation or government, denominated in the currency of the country where it is issued (a British firm's dollar bonds in the USA).

40 **What are fully hedged bonds?**

A Foreign bonds whose currency fluctuation risk is eliminated by selling the entire stream of principal and interest payments in forward markets.

41 **What is a medium term note (MTN) programme?**

A A programme under which several lots of bonds with different features (coupons, currencies) are issued at times chosen for market opportunities, with documentation and regulatory approvals taken at one time.

42 **Name the three kinds of ESG bonds and how they differ.**

A Green bonds and social bonds are project-based (proceeds for specific green or social projects); sustainability-linked bonds are target-based (proceeds for general corporate purposes to achieve KPIs).

43 **What are the features of a drop lock bond?**

A A floating rate note with a normal floating rate, automatically converted into a fixed rate bond if interest rates fall below a predetermined level; the fixed rate stays until maturity; a short option structure.

44 **What are American Depository Receipts?**

A Securities offered by non-US companies that want to list on a US exchange; each ADR represents a certain number of the company's regular shares, letting US investors buy them without investing directly on a foreign exchange.

45 **What are Global Depository Receipts?**

A Negotiable certificates held in the bank of one country representing a specific number of shares of a stock traded on the exchange of another country; used to raise dollars or euros; mainly traded in Europe, particularly London.

46 **What are Indian Depository Receipts?**

A Depository receipts issued in India by foreign companies to raise funds from the Indian capital market, on the same lines as ADRs/GDRs; listed and traded in India like other Indian securities.

47 **Why did Indian companies prefer GDRs to ADRs?**

A A US listing exposes them to a higher level of responsibility in disclosure, costs, liabilities and timing; regulations are more stringent; and they must give full, half-yearly and quarterly accounts in accordance with, or reconciled with, US GAAPs.

48 **What is crowdfunding, and who are its three parties?**

A Raising money for an individual or organisation from a group of people, typically via the internet, in exchange for equity, loans (P2P) or nothing (donation). Parties: the fund raiser, the mediator (platform) and the fund investor.

CHAPTER 4

Cost of Capital

4.1 Cost of debt and of preference capital

1 **Formula: cost of irredeemable debentures?**

A Kd = I(1 – t) ÷ NP, where NP = net proceeds (issue price – flotation cost) or the current market price.

2 **Formula: cost of redeemable debentures by the approximation method (only interest tax-deductible)?**

A Kd = [I(1 – t) + (RV − NP) ÷ n] ÷ [(RV + NP) ÷ 2].

3 **Formula: cost of redeemable debentures when discount on issue and premium on redemption are also tax-deductible?**

A Kd = [I + (RV − NP) ÷ n] × (1 – t) ÷ [(RV + NP) ÷ 2].

4 **ICAI's definition of flotation cost?**

A The sum of the expenditure on a new issue of a security, such as underwriting or brokerage fees, legal and administrative charges, registration fees and printing expenses; it is deducted to arrive at net proceeds.

5 **YTM approach: the relevant cash flows, by year?**

A Year 0: net proceeds (new issue) or current market price (existing debt); years 1 to n: interest net of tax, I(1 – t); year n: redemption value.

6 **When should the approximation formula for Kd not be used?**

A When the difference between RV and NP is very high (e.g. deep discount bonds), and for gradual redemption (amortised bonds); use the YTM / trial-and-error method.

7 **Convertible debentures: which redemption value is used?**

A The higher of the cash value of the debenture and the value of the shares at maturity (shares per debenture × current price × $(1 + g)^n$).

8 **Why is preference dividend not adjusted for tax?**

A It is an appropriation of after-tax profit, not an expense, so it does not reduce the tax liability.

9 **Formula: cost of redeemable preference shares (approximation)?**

A $Kp = [PD + (RV - NP) \div n] \div [(RV + NP) \div 2]$.

10 **What is the tax shield on interest?**

A The saving in tax liability because interest is allowed as an expense: interest × tax rate (₹40 lakh × 35% = ₹14 lakh in ICAI's Example 1).

11 **The five features of debentures or bonds, in ICAI's order?**

A Face value; interest (coupon) rate; maturity period; redemption value; benefit of tax shield.

12 **How is the cost of a zero coupon (deep discount) bond found?**

A By trial and error from $RV = NP \times (1 + r)^n$, since no interest is paid; not by the approximation formula.

4.2 Cost of equity and of retained earnings

13 **Formula: cost of equity by the dividend price approach, and its assumption?**

A $Ke = D \div P0$ (expected dividend ÷ ex-dividend market price); assumes the dividend per share stays constant forever.

14 **Earnings price approach: formula and how it differs from the dividend price approach?**

A Ke = E ÷ P (current EPS ÷ market price), assuming constant EPS forever; it seeks to nullify the effect of changes in the dividend policy.

15 **Gordon's model for the cost of equity, existing and new shares?**

A Ke = D1 ÷ P0 + g; for new shares Ke = D1 ÷ (P0 − F) + g, where F is the flotation cost per share and D1 = D0(1 + g).

16 **Two methods of estimating g?**

A Average method: g = $\sqrt[n]{(D0 \div Dn)} - 1$ (or via the FVIF table); Gordon's growth model: g = b × r (retention rate × return on funds invested).

17 **Formula: CAPM cost of equity?**

A Ke = Rf + β(Rm − Rf); Rm − Rf is the market risk premium.

18 **Which risk does CAPM reward, and why?**

A Only systematic (non-diversifiable) risk, measured by beta; unsystematic risk can be eliminated by diversification.

19 **Two shortcomings of CAPM in the Study Material?**

A Estimation of beta with historical data is unrealistic; market imperfections may lead investors to unsystematic risk.

20 **Realised yield approach: its assumptions that ICAI calls unrealistic?**

A The risks faced by the company remain the same; shareholders continue to expect the same rate of return; their reinvestment opportunity rate equals the realised yield.

21 **What is the cost of retained earnings?**

A The opportunity cost of dividends foregone by shareholders: the return they could have earned by investing in financial assets.

22 **Kr versus Ke: which price, and why is Ke normally higher?**

A Kr uses the current market price with no flotation cost; Ke on a new issue uses net proceeds (issue price less flotation), so Ke is normally higher.

4.3 WACC and the marginal cost of capital

23 ICAI's definition of WACC?

A The weighted average after-tax costs of the individual components of the firm's capital structure, combined into a single overall cost of capital.

24 The four steps to calculate WACC?

A (1) Total capital from all sources; (2) proportion of each source to total capital; (3) multiply each proportion by its cost; (4) add the products.

25 Three decisions in which the cost of capital is significant?

A Evaluation of investment options, financing decisions, and designing an optimum credit policy.

26 The five factors affecting the cost of capital (SM order)?

A Supply and demand, investor preferences, risk and return, inflation, exchange rates.

27 **How are retained earnings given market value weights?**

A The market value of equity shares is apportioned between paid-up equity capital and retained earnings in the ratio of their book values.

28 **Book value versus market value weights: the five points of comparison?**

A Definition (historical cost vs current market price), reflection of current conditions, stability, relevance for investors, ease of calculation.

29 **ICAI's definition of marginal cost of capital?**

A The cost of raising an additional rupee of capital (the cost of new funds), derived as the average cost of capital calculated with marginal weights.

30 **What are marginal weights, and why is there no BV/MV choice for MCC?**

A The proportions of funds the firm intends to employ; since the weights are those of the new funds, book and market values do not arise.

31 **Formula: break point in the marginal cost of capital?**

A Amount of the cheaper source available ÷ its proportion in new funds (e.g. retained earnings ÷ equity proportion).

32 **When is MCC equal to the average cost of capital, and what happens when component costs rise?**

A Equal when funds are raised in the same proportions and component costs stay unchanged; when component costs rise, the average cost rises and the marginal cost rises faster.

33 **A company earns 20% and its WACC is 11%. What does the Study Material conclude?**

A It produces value of about 9% for its investors; a return below WACC means the company is shedding value.

CHAPTER 5

Financing Decisions: Capital Structure

5.1 Capital structure theories: NI, NOI, traditional and Modigliani-Miller

1 **State the seven assumptions ICAI lists for capital structure theories.**

A Only debt and equity; total assets given, leverage changed by swapping debt and shares; no taxes; 100% dividend payout; total financing constant; business risk constant; perpetual life.

2 **NI approach: what is constant, and what does it recommend?**

A Ke and Kd are constant, so more debt lowers WACC and raises value; it pleads for as much debt as possible.

3 **NOI approach: how are value and Ke found?**

A V = EBIT ÷ Ko (Ko constant); S = V − D; Ke = (EBIT − interest) ÷ S, which rises with leverage. Capital structure is irrelevant.

4 **Traditional approach: the three features?**

A Kd constant for a while then rises; Ke constant or gradually rising, then rising speedily; WACC first falls then rises, its lowest point being the optimal capital structure.

5 **MM 1958: the four additional assumptions?**

A Perfect capital markets (free information, no transaction costs); rational investors; equivalent risk classes by business risk; no corporate taxes.

6 **MM without tax: cost of equity of a levered firm?**

A Ke = Ko + (Ko − Kd) × Debt ÷ Equity.

7 **MM with tax: value of a levered firm?**

A Vg = Vu + TB, the present value of tax shields (debt × tax rate for permanent debt); Vu = EBIT (1 − t) ÷ Keu.

8 **MM with tax: Ke and Ko of the levered firm?**

A Keg = Keu + (Keu − Kd) × Debt (1 − t) ÷ Equity; Kog = Keu (1 − tL), L = debt ÷ (debt + equity).

9 **Define arbitrage (ICAI).**

A Buying an asset or security at a lower price in one market and selling it at a higher price in another, so that equilibrium is attained in different markets.

10 **Shortcoming of the MM approach?**

A The arbitrage process fails because of capital market imperfections, transaction costs and corporate income taxes.

11 **Trade-off theory in one line?**

A Firms balance the tax benefit of debt against the costs of financial distress (bankruptcy and non-bankruptcy) and agency costs; this gives an optimal capital structure.

12 **Pecking order theory: the order of financing?**

A Internal finance first; then secured, unsecured and hybrid debt; new equity shares last. Based on asymmetric information.

5.2 EBIT-EPS analysis, indifference

point and financial break-even

13 **Define the financial break-even point (ICAI).**

A The minimum level of EBIT needed to satisfy all the fixed financial charges, i.e. interest and preference dividends; the EBIT at which EPS equals zero.

14 **Formula: financial break-even point?**

A Interest + preference dividend ÷ (1 – tax rate).

15 **What is the indifference point?**

A The EBIT at which EPS is the same under two financing alternatives: the point where their EPS lines intersect on the EBIT-EPS chart (EPS equivalency point).

16 **Formula: indifference point between two plans with debt?**

A $(EBIT - I_1)(1 - t) \div E_1 = (EBIT - I_2)(1 - t) \div E_2$, where E = number of equity shares and I = interest.

17 **Two plans have the same equity share capital. When is there no indifference point?**

A When the after-tax cost of the other source differs between the plans: one plan is better at every EBIT.

18 **Two plans have the same equity share capital and the same after-tax cost of the other source. What follows?**

A Every EBIT is an indifference point.

19 **Why is debt generally preferred to preference shares in EBIT-EPS analysis?**

A Interest is usually lower than the preference dividend rate, and interest is tax-deductible, so its after-tax cost is lower.

20 **When does fixed-charge financing increase EPS?**

A When the rate of return on assets is higher than the cost of the fixed-charge funds: favourable financial leverage, or trading on equity.

21 **Plans have different PE ratios. Which plan is chosen?**

A The one with the highest market price per share (EPS × PE ratio), even if its EPS is lower.

22 **Number of new shares when shares of ₹10 are issued at a premium of ₹15?**

A Amount raised ÷ ₹25, the issue price.

23 **ICAI's three analyses for choosing an optimal capital structure?**

A EBIT-EPS-MPS analysis, indifference point analysis and financial break-even point analysis.

5.3 Capital structure planning, determinants and over- and under-capitalisation

24 **Define capital structure (ICAI).**

A The combination of capitals from different sources of finance: equity shareholders' fund, preference share capital and long-term external debts.

25 **The three factors keeping in mind which the source and quantum of capital are decided?**

A Control (existing shareholders keep majority stake), risk (financial risk within tolerable limits), cost (overall cost of capital minimum).

26 **What does the capital structure decision decide?**

A The forms of financing (sources), their actual requirements (amounts) and their relative proportions (mix) in total capitalisation.

27 **List ICAI's seven principles (factors) of capital structure, in order.**

A Financial leverage or trading on equity; growth and stability of sales; cost principle; risk principle; control principle; flexibility principle; other considerations (nature of industry, timing of issue, competition).

28 **Define trading on equity.**

A The use of long-term fixed interest bearing debt and preference share capital along with equity share capital; it raises EPS when the firm earns more than the cost of debt.

29 **The three analyses for an optimal capital structure?**

A EBIT-EPS-MPS analysis, indifference point analysis, financial break-even point analysis.

30 **Define over-capitalisation and give its chief sign.**

A A firm has more capital than it needs, or assets are worth less than its issued share capital, and earnings are insufficient to pay dividend and interest. Chief sign: fall in payment of dividend and interest, and in the value of shares.

31 **Five causes of over-capitalisation?**

A Raising more than can be employed profitably; borrowing at a rate higher than the company earns; excessive payment for fictitious assets such as goodwill; improper provision for depreciation and replacement and high dividends; wrong estimation of earnings and capitalisation.

32 **Four consequences of over-capitalisation?**

A Considerable reduction in rate of dividend and interest; reduction in market price of shares; window dressing; reorganisation, or even liquidation.

33 **Four remedies for over-capitalisation?**

A Thorough reorganisation; buyback of shares; reduction in claims of debenture-holders and creditors; reduction in the value of shares.

34 **Define under-capitalisation.**

A A state when actual capitalisation is lower than proper capitalisation as warranted by earning capacity; typical of companies with large secret reserves not brought into the books.

35 **Three remedies for under-capitalisation?**

A Split up the shares; issue bonus shares (the most appropriate measure); revise upward the par value of shares in exchange for existing shares.

CHAPTER 6

Financing Decisions: Leverages

6.1 Operating, financial and combined leverage

1 **Define business risk and financial risk (ICAI).**

A Business risk: the risk associated with the firm's operations, i.e. uncertainty about future operating income (EBIT). Financial risk: the additional risk placed on shareholders because of use of debt.

2 **Operating leverage is a function of which three factors?**

A Amount of fixed cost, variable contribution margin and volume of sales.

3 **Three formulas for DOL?**

A % change in EBIT ÷ % change in sales; contribution ÷ EBIT; 1 ÷ margin of safety ratio.

4 **Define financial leverage (ICAI).**

A The use of funds with a fixed cost in order to increase earnings per share.

5 **Formula: DFL with preference shares?**

A EBIT ÷ [EBIT − interest − preference dividend ÷ (1 − t)]; without preference shares, EBIT ÷ EBT.

6 **Formula: degree of combined leverage, three ways?**

A DOL × DFL; contribution ÷ EBT; % change in EPS ÷ % change in sales.

7 **DOL when there is no fixed cost, at break-even, and below break-even?**

A 1 (no operating leverage); undefined (infinite); negative.

8 **Can DOL or DFL lie between 0 and 1?**

A No. Each is zero or less, or one or more.

9 **What is trading on equity?**

A The use of funds with fixed cost, like long-term debt and preference share capital, along with equity share capital to increase earnings for equity shareholders; favourable when earnings exceed the cost of debt.

10 **Why is financial leverage called a double-edged sword?**

A When the cost of fixed cost funds is below the return on investment it raises ROE and EPS; when above, it reduces them and can lead to financial distress.

11 **DOL-DFL combinations: best and riskiest?**

A Best: low DOL, high DFL (moderate risk). Riskiest: high DOL, high DFL.

12 **A 20% fall in sales would wipe out EPS. What is the DCL?**

A 100% ÷ 20% = 5.

CHAPTER 7

Investment Decisions

7.1 Capital budgeting and estimating relevant cash flows

1 **Capital budgeting decisions are critical for four reasons. Name them in ICAI's order.**

A Substantial investment; long time period; irreversibility; complex decisions.

2 **The six phases of the capital budgeting process, in order?**

A Planning, evaluation, selection, implementation, control, review.

3 **Types of capital investment decisions on the basis of the firm's existence?**

A Replacement and modernisation (cost reduction decisions); expansion and diversification (revenue expansion decisions).

4 **Types of capital investment decisions on the basis of situations?**

A Mutually exclusive, accept-reject and contingent decisions.

5 **Which three factors determine the life of a project?**

A Technological obsolescence, physical deterioration and decline in demand for the product.

6 **ICAI's definition of sunk cost, and its treatment?**

A An outlay of cash already incurred in the past that cannot be reversed in present; excluded from capital budgeting analysis.

7 **ICAI's definition of opportunity cost, and its treatment?**

A Foregoing of a benefit due to choosing an alternative investment option; included in cash outflows, at the start or during the project.

8 **How is working capital treated in project cash flows?**

A Initial requirement: outflow; additional need: outflow when needed; reduction: inflow; release at the end: inflow, in full if nothing is said; no depreciation on it.

9 **Exclusion of financing costs principle: what is excluded and why?**

A Interest on long-term debt and dividends; they are reflected in the WACC, so deducting them from cash flows counts the cost of funds twice.

10 **Formula: profit after tax excluding financing cost?**

A PBIT × (1 – t), which equals PAT + interest × (1 – t).

11 **Terminal year, asset is the only one in its block: treatment?**

A No depreciation in that year; opening WDV – sale value is a short-term capital loss (tax benefit) or, if negative, a short-term capital gain (tax outflow).

12 **Terminal year, other assets remain in the block: treatment?**

A Depreciation on (opening WDV – sale value) at the block rate; none if the sale value exceeds the opening WDV.

7.2 Payback, discounted payback and ARR

13 **ICAI's definition of payback period?**

A Time required to recover the initial cash-outflow; the length of time for the cumulative net cash flows to equal the total initial cash outlays.

14 **Payback period with non-uniform cash flows: how is the fraction of the last year found?**

A Unrecovered outlay at the start of that year ÷ cash inflow of that year.

15 **Three advantages of the payback period (ICAI)?**

A Easy to compute; easy to understand as a quick estimate of the time to recoup the cash invested; its length can serve as an estimate of risk.

16 **Three limitations of the payback period (ICAI)?**

A Ignores the time value of money; ignores total profitability (cash flows after payback); emphasis on short payback ignores long-term projects.

17 **Formula and conditions for the payback reciprocal?**

A Average annual cash inflow ÷ initial investment; a close approximation of IRR if the life is at least twice the payback period and annual cash inflows are equal.

18 **ICAI's definition of ARR?**

A The average annual net income of the project (incremental income) as a percentage of the investment (initial or average).

19 **Formula: average investment for ARR?**

A ½ (initial investment − salvage value) + salvage value, plus additional working capital if any.

20 **Four limitations of ARR (ICAI)?**

A Ignores the time value of money; depends on accounting procedures such as depreciation; ignores cash flows; book value ignores working capital and other outlays.

21 **What is the discounted payback period, and why is it considered superior to simple payback?**

A The payback period computed on cash flows discounted at a predetermined rate; it takes the time value of money into account.

22 **As the required rate of return increases, what happens to the gap between simple and discounted payback?**

A The distortion between them grows.

23 **Decision rule for payback and ARR, independent and mutually exclusive projects?**

A Payback: accept if ≤ maximum acceptable payback; choose the least payback. ARR: accept if ≥ minimum acceptable rate; choose the maximum ARR.

7.3 NPV, profitability index and equivalent annual cost

24 **ICAI's definition of net present value?**

A The amount, in current value of amount, the investment earns after paying cost of capital in each period.

25 **Formula: NPV when there are outflows during the project's life?**

A Present value of cash inflows – present value of cash outflows.

26 **The five steps for calculating NPV (ICAI)?**

A Determine each year's net cash inflow; select the discount rate (WACC); find each year's discount factor; multiply cash flows by the factors; total the present values.

27 **NPV decision rule for independent and mutually exclusive projects?**

A Accept if NPV ≥ 0, reject if NPV < 0; among mutually exclusive projects, the highest positive NPV.

28 **Three limitations of NPV (ICAI)?**

A Difficult calculations; accuracy depends on forecasting cash flows and the discount rate; an absolute measure that ignores differences in initial outflow and size.

29 **Profitability index: formula and its two other names?**

A Sum of discounted cash inflows ÷ initial cash outlay or total discounted cash outflow; also called desirability factor and present value index.

30 **PI decision rule for independent and mutually exclusive projects (ICAI's summary)?**

A Accept if PI ≥ 1, reject if PI < 1; among mutually exclusive projects, when NPV is the same, the highest PI.

31 **Three limitations of the profitability index (ICAI)?**

A Fails in capital rationing with indivisible projects; choosing one large high-NPV project can exclude several small ones with a higher total NPV; a lower-PI project may release cash for another project later, for a higher total NPV.

32 **Two methods for mutually exclusive projects with unequal lives?**

A Replacement chain (common life) method and equivalent annualised criterion.

33 **The three steps of the equivalent annualised criterion?**

A Compute NPV at the WACC; compute the PVAF at that rate for each project's life; divide NPV by PVAF and compare.

34 **Equivalent annual cost: formula and decision?**

A PV of costs over the life (net of salvage) ÷ PVAF for that life; choose the lower equivalent annual cost.

7.4 IRR, MIRR and the NPV-IRR conflict

35 **ICAI's definition of internal rate of return?**

A The discount rate that equates the present value of the expected cash inflows with the initial cash outflow.

36 **Formula: IRR by interpolation?**

A LR + [NPV at LR ÷ (NPV at LR − NPV at HR)] × (HR − LR), where LR and HR are the lower and higher trial rates.

37 **IRR with uniform cash inflows: the first step?**

A Compute the fake (approximate) payback period, investment ÷ annual cash inflow, and look it up in the PVAF table in the row for the project's life.

38 **IRR acceptance rule?**

A Accept if IRR ≥ cut-off rate or WACC; reject if IRR < cut-off rate; indifferent if equal.

39 **Reinvestment assumption of NPV and of IRR?**

A NPV: cash flows reinvested at the discount rate used for NPV. IRR: cash flows reinvested at the project's IRR, so IRR favours projects with heavy early cash flows.

40 **When do multiple IRRs arise, and when should such a project be accepted?**

A When cash flows change signs during the life (for example a major outflow at the end); accept only if the cost of capital lies between IRR_1 and IRR_2.

41 **Three scenarios in which NPV and IRR give different conclusions?**

A Scale or size disparity; time disparity in cash flows; disparity in lives (unequal lives).

42 **NPV and IRR conflict for mutually exclusive projects: which is followed, and why?**

A NPV: it generally gives consistent results in conformity with the wealth maximisation principle (it measures the absolute addition to shareholders' wealth).

43 **What is MIRR, and what is its other name?**

A The rate that equates the PV of the terminal value of all inflows (compounded at usually the cost of capital) with the year-0 outflow; also called the terminal value method.

44 **Formula: MIRR?**

A The nth root of (terminal value of the inflows ÷ initial outflow), minus 1; n = life in years.

45 **Three deficiencies of IRR that MIRR addresses (ICAI)?**

A It eliminates multiple IRR rates; it addresses the reinvestment rate issue; it produces results consistent with the NPV method.

46 **Four advantages of IRR (ICAI)?**

A Uses the time value of money; considers all cash flows; easier to use, as comparison with the cost of capital gives instant understanding of desirability; helps achieve maximisation of shareholders' wealth.

7.5 Capital rationing

47 **What is capital rationing (ICAI)?**

A A situation where funds are restricted and the management has to choose from amongst available alternative investments.

48 **Capital rationing: how are independent, divisible projects ranked?**

A On the basis of NPV per rupee of capital (NPV ÷ investment); accept in rank order, the last project in part.

49 **Capital rationing: how are indivisible projects selected?**

A Rank on absolute NPV, examine the combinations within the funds available, and select the combination with the highest total NPV.

50 **Relation between NPV per ₹1 and the profitability index?**

A NPV per ₹1 = PI − 1 when the whole outlay is at year 0, so both rank projects the same way.

51 **Which limitation of PI matters under capital rationing?**

A It fails as a guide in resolving capital rationing where projects are indivisible.

52 **Illustration 13 (Shiva Ltd): why does E + B + D beat E + B + C if projects are indivisible?**

A Its total NPV is higher (₹49.2 thousand against ₹48.1 thousand), despite D ranking lower than C on NPV per rupee.

53 **Under capital rationing, may a negative-NPV project be accepted to use up idle funds?**

A No: it reduces shareholders' wealth; it is rejected with or without rationing.

54 **NPV of a project taken up in part?**

A Its NPV × (funds remaining ÷ its investment).

CHAPTER 8

Dividend Decision

8.1 Forms of dividend, buy-back and determinants of dividend policy

1 **ICAI's meaning of dividend?**

A That part of profit after tax which is distributed to the shareholders of the company; the rest is retained as surplus for future growth.

2 **What is the ex-dividend price?**

A The price at which shares trade immediately on the next day after declaration of dividend, when they no longer carry the right to the dividend.

3 **Three forms of dividend (ICAI)?**

A Cash dividend; share repurchases (treasury or cancelled shares); stock dividend (bonus shares).

4 **SEBI conditions for a bonus issue (four)?**

A Not in lieu of cash dividend; authorised by the AOA and all partly paid shares made fully paid; no default in repaying loans, interest or statutory dues; issued only from share premium and free reserves, not from revaluation reserve.

5 **Two advantages of a stock dividend to the company?**

A Conserves cash for profitable investment opportunities; suitable in case of cash deficiency and restrictions imposed by lenders on cash dividends.

6 **Stock split: meaning, two advantages, two limitations?**

A One share split into many (₹500 into 5 × ₹100). Advantages: affordable to small investors; more shareholders. Limitations: additional expenditure; low price may attract speculators or short-term investors.

7 **Two routes for a share buy-back?**

A Open market (through the secondary market) and tender offer (a fixed price at which all shareholders can sell).

8 **Formula linking retention and growth?**

A g = b × r, where b = retention ratio and r = rate of return on investment.

9 **The nine determinants of dividend decisions, in ICAI's order?**

A Availability of funds, cost of capital, capital structure, stock price, investment opportunities in hand, trend of industry, expectation of shareholders, legal constraints, taxation.

10 **Lintner's model: formula and its two parameters?**

A $D_1 = D_0 + [(EPS \times \text{target payout}) - D_0] \times Af$; parameters: the target payout ratio and the speed at which current dividends adjust to the target.

11 **Two limitations of Lintner's model?**

A It does not offer a market price for the shares; the adjustment factor is arbitrary, not based on any scientific criterion or method.

12 **Mature versus growth companies: payout ratios?**

A Mature companies with few investment opportunities show high payout ratios; growth companies, needing funds for expansion, show low payout ratios.

8.2 Dividend theories: traditional,

Walter, Gordon and Modigliani-Miller

13 **Walter's model: formula?**

A $P = [D + (r \div Ke)(E - D)] \div Ke$, where D = DPS, E = EPS, r = return on investment, Ke = cost of equity (capitalisation rate).

14 **Walter's model: optimum payout for growth, constant and declining firms?**

A Growth (r > Ke): zero. Constant (r = Ke): every payout ratio is optimum. Declining (r < Ke): 100%.

15 **Six assumptions of Walter's model?**

A Investments financed by retained earnings only; r and Ke constant; perfect capital markets; no taxes or tax discrimination; no floatation or transaction cost; perpetual life.

16 **Gordon's model: formula (earnings form)?**

A $P_0 = E_1(1 - b) \div (Ke - br)$, where b = retention ratio and br = g.

17 **Gordon's model: optimum payout when r > Ke, r = Ke, r < Ke?**

A Zero; there is no optimum ratio; 100%.

18 **Seven assumptions of Gordon's model?**

A All-equity firm; IRR constant; Ke constant; retention ratio constant; growth (br) constant; Ke > g; investments financed by retained earnings only.

19 **Bird-in-hand theory (Gordon's revised model): its two arguments?**

A Investors are risk averse; they put a premium on a certain return and a discount on an uncertain return, so they prefer current dividends.

20 **Dividend discount model: zero-growth and constant-growth price?**

A Zero growth: $P_o = D \div Ke$. Constant growth: $P_o = D_o(1 + g) \div (Ke - g)$.

21 **MM hypothesis: the central claim?**

A The market value of equity shares depends solely on the firm's earning power, not on how earnings are split between dividends and retained earnings; dividend policy is irrelevant.

22 **Five assumptions of the MM dividend hypothesis?**

A Perfect capital markets; no taxes (or no tax discrimination); fixed investment policy (equity financing); no floatation or transaction cost; no risk of uncertainty.

23 **MM: formulas for P_0 and for the value of the firm?**

A $P_0 = (P_1 + D_1) \div (1 + Ke)$; $nP_0 = [(n + \Delta n)P_1 - I + E] \div (1 + Ke)$, with $\Delta n = [I - (E - nD_1)] \div P_1$.

24 **Graham and Dodd's model?**

A $P = m(D + E/3)$: the market places considerable weight on dividends than on retained earnings; m is a multiplier.

CHAPTER 9

Management of Working Capital

9.1 Working capital, the operating cycle and financing policy

1 **ICAI's definition of the working capital (operating) cycle?**

A The length of time between a company's paying for materials, entering into stock and receiving the cash from sales of finished goods; found by adding the days required for each stage.

2 **Formula: operating cycle, with each letter named?**

A R + W + F + D − C: raw material storage period, WIP holding period, finished goods storage period, receivables collection period, minus credit period allowed by suppliers.

3 **Denominators of R, W, F, D and C?**

A R: average raw material consumption per day; W: average cost of production per day; F: average cost of goods sold per day; D: average credit sales per day; C: average credit purchases per day.

4 **Gross working capital and net working capital?**

A Gross: the firm's investment in current assets. Net: current assets minus current liabilities.

5 **Permanent working capital (ICAI)?**

A The base working capital: the minimum level of investment in current assets carried at all times to carry on day-to-day activities; generally financed by long-term sources.

6 **Temporary working capital (ICAI)?**

A The part of total working capital required in addition to permanent working capital, due to fluctuation in sales volume (e.g. seasonal inventory); also called variable or fluctuating; may be financed by short-term sources.

7 **Thumb rules ICAI quotes for a good working capital position?**

A Current ratio of 2 for a manufacturing firm and quick (acid test) ratio of at least 1; the true optimum depends on the circumstances of the business.

8 **The 3 Es of working capital management?**

A Economy in financing, efficiency in utilisation and effectiveness in achieving the intended objectives.

9 **Aggressive vs conservative working capital investment: liquidity, risk and the CA/FA ratio?**

A Aggressive: minimal current assets, poor liquidity, higher risk and return, lower CA/FA ratio. Conservative: high current assets, greater liquidity, lower risk and return, higher CA/FA ratio.

10 **The six parameters that guide the choice of working capital finance?**

A Cost factor; impact on credit rating; feasibility; reliability; restrictions; hedging or matching approach (financing assets with funds of the same maturity).

11 **Spontaneous vs negotiated sources of working capital finance?**

A Spontaneous arise naturally in the course of business (trade credit, credit from employees and service suppliers); negotiated must be specifically negotiated with banks, financial institutions or the public.

9.2 Estimating working capital requirement

12 Three methods of estimating working capital needs (ICAI)?

A Current assets holding period (the operating cycle concept), ratio of sales, and ratio of fixed investments.

13 Formula: working capital by the operating cycle method?

A Annual operating cost ÷ number of operating cycles in the year (on cash cost basis, exclude depreciation from the operating cost).

14 Formula for estimated receivables in the working capital statement (SM 9.23)?

A Estimated credit sales (units) ÷ 12 months (365 days) × estimated cost of sales (excluding depreciation) per unit × average receivable collection period.

15 Why are debtors and finished goods taken at cash cost?

A Sales to debtors include cost and profit, and the funds required need not include profit; non-cash expenses like depreciation are also excluded. Only the funds actually blocked are financed.

16 **Which current liabilities are deducted in the estimate?**

A Current liabilities other than bank credit arising in the normal course of business: trade payables, wages payable, overheads payable.

17 **Work-in-progress when raw materials are fed in at the start and labour and overheads accrue evenly?**

A Full raw material for the period in process plus half of the labour and overheads for that period.

18 **Cash cost of production and cash cost of sales (Illustration 4)?**

A Cash cost of production = materials + wages + cash manufacturing expenses. Cash cost of sales = cash cost of production + administrative expenses + selling (sales promotion) expenses.

19 **ICAI's statement of working capital, the order of the totals?**

A I Current assets (gross working capital) – II Current liabilities = III Excess of current assets over current liabilities; + IV Safety margin = V Net working capital.

20 **Effect of double shift working on work-in-progress?**

A The amount of materials in process does not change, since work started in the first shift is completed in the second; WIP cost changes only if second-shift workers are paid more (or cost per unit changes).

21 **Why may stocks not double when double shift doubles production?**

A The minimum level of stocks may not be much higher, so the increase in stocks may not be proportionate to the rise in production.

9.3 Management of receivables, credit policy, factoring and forfaiting

22 **The basic objective of management of receivables (ICAI)?**

A To optimise the return on investment on these assets.

23 **The three aspects of management of receivables?**

A Credit policy (credit standards, credit terms, collection efforts), credit analysis, and control of receivables.

24 **What does "3/15 net 60" mean?**

A A 3% cash discount if the customer pays within 15 days; if the offer is not availed, payment must be made within 60 days.

25 **ICAI's four costs of maintaining receivables?**

A Cost of additional funds blocked (interest on loan funds or opportunity cost of own funds), administrative costs, collection costs and defaulting costs.

26 **Formula: opportunity cost of investment in receivables (ICAI format)?**

A Total cost of credit sales × collection period (days) ÷ 365 (or 360) × required rate of return ÷ 100.

27 **Total fixed cost in a credit policy question, from average cost?**

A (Average cost per unit − variable cost per unit) × number of units sold on credit under the present policy.

28 **Pledging vs factoring of receivables?**

A Pledging uses receivables as security for a short-term loan (50–90% financed; adds debt). Factoring is the outright sale of receivables to a factor, which bears the collection risk and services the accounts for a fee; it is not a loan.

29 **Factoring with recourse and without recourse?**

A With recourse: if the factor cannot collect, it turns the receivable back to the firm (generally replaced with new receivables). Without recourse: the factor bears the ultimate risk of default and charges a higher commission.

30 **Rate of effective cost of factoring?**

A Net annual cost of factoring ÷ advances to be paid × 100 (or ÷ amount available for advance); advances to be paid = amount available for advance − interest deducted by the factor.

31 **When should a firm avail factoring (ICAI's advice)?**

A If the rate of effective cost of factoring is less than the existing cost of borrowing, or if factoring results in positive net annual benefits; not if the effective cost exceeds the cost of borrowing.

32 **Forfaiting: meaning and recourse?**

A From the French "relinquish a right": a bank or financial institution buys exporters' trade bills or receivables, the exporter relinquishing the right to receive payment from the importer; always without recourse.

33 **The five steps of the debt collection programme?**

A Monitoring the state of receivables; intimation as the due date approaches; e-mail and telephonic advice on the due date; reminding the legal recourse on overdue accounts and following an escalation matrix; legal action on overdue accounts.

9.4 Treasury and cash management

34 **Define treasury management (ICAI).**

A Planning, organizing and controlling the funds and working capital of an enterprise in order to ensure best use of funds, maintain liquidity, reduce overall cost of funds and mitigate operational and financial risk.

35 **The five fundamental tasks of the treasury department, in ICAI's order?**

A Cash management, currency management, fund management, banking, corporate finance.

36 **The three main objectives of cash management?**

A Provide adequate cash to each unit as per requirements; no funds blocked in idle cash; surplus cash invested to maximise returns.

37 **The four kinds of float, in order?**

A Billing float (sale to mailing of invoice), mail float (cheque in transit), cheque processing float (receipt to deposit), banking processing float (deposit to credit of funds).

38 **Lock box system: main advantage and main drawback?**

A Advantage: cheques are deposited sooner, eliminating the cheque processing float. Drawback: cost, almost proportional to the number of cheques, so not profitable when the average remittance is small.

39 **ICAI's rule for deciding on concentration banking or a lock box?**

A Compare the added cost of the most efficient system with the marginal income from the released funds; profitable if the cost is less than the income.

40 **Baumol model: formula and symbols?**

A $C = \sqrt{(2U \times P \div S)}$; C = optimum cash balance, U = annual (or monthly) cash disbursement, P = fixed cost per transaction, S = opportunity cost of one rupee p.a. (or p.m.).

41 **Baumol model: the four assumptions?**

A Cash needs known with certainty; cash used uniformly and known with certainty; holding cost known and constant; transaction cost constant.

42 **Miller-Orr: what happens at the upper limit, at the lower limit, and in between?**

A At h, invest h − z in marketable securities; at the lower limit, transfer from marketable securities to cash; between the limits, no transactions.

43 **Miller-Orr: the limits are set on the basis of which three factors?**

A The fixed cost of securities transactions, the opportunity cost of holding cash and the degree of likely fluctuations in cash balances.

44 **Three principles for selecting marketable securities?**

A Safety, maturity, marketability.

45 **Advantages of virtual banking (ICAI)?**

A Lower cost of handling a transaction; increased speed of response to customers; lower cost of operating the branch network and reduced staff costs; improved range of services available rapidly, accurately and at the customer's convenience.

9.5 Management of inventory

46 **The seven problems inventory management covers, in ICAI's order (FM Study Material 9.69)?**

A Fixation of minimum and maximum levels; determining the size of inventory to be carried; deciding the issues, receipts and inspection procedures; determining the EOQ; proper storage facilities; keeping check over obsolescence; ensuring control over movement of inventories.

47 **Inventory row of ICAI's working capital trade-off table?**

A Higher side: fewer stock-outs increase profitability. Trade-off: use EOQ, JIT etc. to carry the optimum level. Lower side: less capital required, but endangers stock-out and loss of goodwill.

48 **Define inventory control (CIMA, quoted by ICAI).**

A The function of ensuring that sufficient goods are retained in stock to meet all requirements without carrying unnecessarily large stocks.

49 **EOQ formula and its symbols?**

A $\sqrt{(2 \times A \times O \div C)}$: A = annual requirement, O = cost per order, C = carrying cost per unit per annum.

50 **Carrying costs: what do they include (ICAI)?**

A Cost of fund invested in inventories, cost of storage, insurance cost, obsolescence.

51 **Average stock with an order quantity Q and a safety stock?**

A Q ÷ 2 + safety stock; the money invested is that average × cost per unit.

52 **JIT: definition and its two principles?**

A A system with an approach to have zero inventories in stores, purchasing material only when actually required for production. Principles: produce goods only when required; deliver products to customers only when they want them.

53 **Raw material storage period formula (SM 9.20)?**

A Average stock of raw material ÷ average cost of raw material consumption per day.

54 **"Desired level of inventory" as a factor in planning working capital?**

A The level that allows uninterrupted production but reduces the investment in raw materials and hence increases cash flow; JIT and EOQ are used for it.

55 **How does ICAI say inventory is ideally financed?**

A By credit granted by the supplier; depending on the cash conversion cycle, a bank loan or overdraft, or factoring of debtors, may be needed.

9.6 Management of payables

56 **Why is trade credit called a spontaneous source of finance?**

A Because it arises from ordinary business transactions, naturally, in the course of business operations.

57 **ICAI's four costs of availing trade credit?**

A Price (the discount forgone, a high implicit cost), loss of goodwill, cost of managing, conditions (minimum order size or regular orders).

58 **ICAI's three costs of not taking trade credit?**

A Impact of inflation, interest (trade credit is an interest free loan), inconvenience to a supplier geared to deferred payment.

59 **Formula: nominal annual cost of not taking a cash discount?**

A d ÷ (100 − d) × 365 ÷ t, where d = discount % and t = days credit outstanding − discount period.

60 **Formula: cost of lost cash discount with compounding?**

A (100 ÷ (100 − d))^(365/t) − 1.

61 **In the discount formulas, what is t?**

A The reduction in the payment period, in days, necessary to obtain the early discount: days credit outstanding − discount period.

62 **Illustration 18: what is wrong with comparing a 2% discount with a 12% bank rate?**

A The 2% is earned for paying a few days earlier, the 12% is for a year; the discount must be annualised before comparing.

63 **If a firm cannot borrow and must rely on trade credit, when should it pay?**

A On the last day of the credit period, to reduce the annual cost of the forgone discount.

64 **What does "3/15 net 60" mean?**

A A 3% discount if payment is made within 15 days; otherwise full payment within 60 days.

65 **About what share of total short-term requirements does trade credit contribute, per ICAI?**

A About one-third.

9.7 Financing of working

capital and bank credit

66 **How should permanent and temporary working capital be financed (ICAI)?**

A Permanent working capital by long-term sources such as debt and equity; temporary working capital by short-term sources.

67 **Spontaneous vs negotiated sources of working capital finance?**

A Spontaneous sources arise naturally in the course of business (trade credit, credit from employees and service suppliers, bills payable, accrued expenses); negotiated sources have to be specifically negotiated with lenders such as banks, financial institutions and the public.

68 **The six parameters guiding the choice of a working capital source?**

A Cost factor; impact on credit rating; feasibility; reliability; restrictions; hedging or matching approach (same maturity as the assets).

69 **Commercial paper: maturity and denomination per the Study Material?**

A Minimum 7 days to less than 1 year from the date of issue; ₹5 lakh or multiples thereof.

70 **Commercial paper: its two limitations?**

A Only highly rated firms can use it; it can neither be redeemed before maturity nor extended beyond maturity.

71 **When did RBI withdraw MPBF, and what replaced it?**

A April 1997; banks evolve their own methods of assessing working capital, with Board approval, within prudential guidelines and exposure norms.

72 **The seven forms of bank credit, in ICAI's order?**

A Cash credit, bank overdraft, bills discounting, bills acceptance, line of credit, letter of credit, bank guarantee.

73 **Define a line of credit.**

A A commitment by a bank to lend a certain amount of funds on demand, specifying the maximum amount.

74 **Define a bank guarantee.**

A A facility commercial banks extend on behalf of their clients in favour of third parties, who are the beneficiaries of the guarantees.

75 **Effective cost of a loan with a compensating balance?**

A Interest rate ÷ (1 – minimum balance fraction), counting only a balance the firm would not otherwise maintain; e.g. 16.5% ÷ 90% = 18.33%.

76 **Who introduced the bill rediscounting scheme, and from when?**

A The Reserve Bank of India, with effect from 1 November 1970; all licensed scheduled banks may offer eligible bills to RBI for rediscount.

77 **Who coined the term "core current assets", per the Study Material's MCQ?**

A The Tandon Committee.